

1. (16 points)

A cut in an undirected graph is a separation of the vertices V into two disjoint subsets S and T . The size of a cut is the number of edges that have one endpoint in S and the other in T . Let

$$\mathbf{MAX-CUT} = \{ \langle G, k \rangle \mid G \text{ has a cut of size } k \text{ or more} \}.$$

- (6 points) Give a randomized algorithm that, given a graph G such that G has a cut of size k or more, outputs a cut of expected size at least $k/2$.
- (10 points) Show that **MAX-CUT** is NP-complete. (Hint: Show that $\neq\text{SAT} \leq_P \mathbf{MAX-CUT}$).

2. (16 points) A language is called unary if every string in it is of the form 1^i (the string of i ones) for some $i > 0$. Show that if there exists an **NP**-complete unary language then **P** = **NP**.

Hint: If there is a n^c time reduction from 3SAT to a unary language L , then this reduction can only map size n instances of 3SAT to some string of the form 1^i where $i \leq n^c$. Use this observation to obtain a polynomial-time algorithm for SAT using the downward self reducibility argument of Theorem 2.18 in our textbook.

3. (16 points) Prove that the following language SPACETM is PSPACE-complete:

$$\{ \langle M, w, 1^n \rangle : \text{DTM } M \text{ accepts } w \text{ in space } n \}.$$

4. (16 points) The class DP is defined as the set of languages L for which there are two languages $L_1 \in \text{NP}$, $L_2 \in \text{coNP}$ such that $L = L_1 \cap L_2$. (Do not confuse DP with $\text{NP} \cap \text{coNP}$, which may seem superficially similar.) Show that

- (6 points) EXACT INDSET $\in$ DP.
- (10 points) Every language in DP is polynomial-time reducible to EXACT INDSET.

5. (16 points) Show that the following language is NL-complete:

$$\{ \langle G \rangle : G \text{ is a strongly connected digraph} \}.$$

6. (20 points) Prove that in the certificate definition of NL, if we allow the verifier machine to move its head back and forth on the certificate, then the class being defined changes to NP.